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Visual Analytics for Formula 1 Race Strategies

The 2024 Italian Grand Prix Case Study

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1 Introduction

This report presents our project for the Visual Analytics course held at Sapienza University of Rome during the academic year 2025/2026. The analysis of performance and strategy in Formula 1 has become increasingly complex due to the massive volume of telemetry and timing data generated during a Grand Prix. Modern race strategies are no longer dictated solely by raw speed but by a delicate balance between tyre degradation, fuel loads, and pit-stop efficiency. Visual Analytics plays a crucial role in this domain, transforming high-dimensional data into actionable insights for technical users. This project presents a web-based Visual Analytics solution centered on the 2024 Italian Grand Prix at Monza. It features a comprehensive dashboard developed to explore and analyze multifaceted race dynamics. The system aims to support Race Strategy Engineers in evaluating the efficiency of strategic choices and their direct impact on overall performance, while also providing Sports Journalists with the necessary tools to craft data-driven technical narratives.

2 Related Work

The design of a Visual Analytics system for Formula 1 must be grounded in an understanding of how data visualization can bridge the gap between complex telemetry and strategic decision-making.

2.1 General Frameworks for Formula 1 Visualization

Lamprecht et al. [2] developed an interactive web-based tool providing both a yearly calendar overview and detailed lap-time comparisons. Their work establishes the usefulness of dynamic race position diagrams to reveal racing patterns. While Lamprecht et al. focus on a broad seasonal overview, our project provides a deeper tactical dive into a single race (Monza 2024). We move beyond their standard line plots by integrating a "Strategy Gantt" that encodes tyre life through luminance, and a PCA view to handle the multidimensionality of stints that their tool leaves unexplored.

2.2 Web-Based Multi-Level Analysis

Eckl et al.[1] proposed a framework for analyzing F1 seasons with synchronized views, emphasizing accessibility for both experts and fans. Their system allows for filtering by driver and season to maintain context across different granularities. The primary difference lies in the analytical depth of the interaction. While Eckl's system is primarily descriptive, our dashboard is "white-box" and exploratory. We specifically address the lack of automated analysis in their tool by integrating real-time linear regressions for tyre degradation, triggered directly by user selection.

2.3 Tyre Performance and Strategic Modeling

The tactical importance of tyre management is a central theme in F1 research. Mugge and Zandieh [4] explored using Python and the FastF1 library to model degradation and identify the "optimal window lap life." Their study highlights how models reduce human error in strategy updates. Mugge and Zandieh focus on the predictive side (backend modeling), whereas our project focuses on the interpretative side. Our tool allows a Race Engineer to visually validate these degradation slopes by overlaying regression lines on the Lap Time Evolution chart, transforming static models into interactive visual evidence.

2.4 Tactical Insights and Telemetry

Msakamali [3] investigated the complexities of analyzing team-specific tactical implementations through telemetry and tyre performance case studies. The study underscores that data analysis is essential for distinguishing between varying team strategies during a race. While Msakamali provides a static analysis of these metrics, our solution provides a highly coordinated environment. Our "Parallel Coordinates" view allows for the analysis of technical deltas (Speed Trap, Sector times) that are often treated separately in Msakamali's work, enabling the user to find correlations between high-speed performance and tyre wear.

2.5 Integration of Dimensionality Reduction

Visualizing the high-dimensional nature of a race stint (pace, sectors, temperatures, speeds) is a major challenge. Following the methodology suggested by Sacha et al.[6] for tight integration of automated methods, we adopted Principal Component Analysis (PCA). Unlike the static PCA plots often found in race reports, our PCA view is a dynamic "entry point" for exploration solving the "black box" issue of high-dimensional projections.

3 Dataset and Domain

3.1 Domain Description

The project focuses on the strategic landscape of the 2024 Italian Grand Prix at Monza. This race was selected as a primary case study due to the significant strategic divergence between the top finishers, specifically the contrast between a one-stop and a two-stop strategy. The domain goal is to provide a technical tool for Race Strategy Engineers to evaluate tyre life and performance, and for Sports Journalists to perform post-race technical analysis. The system aims to identify how different tyre compounds and degradation patterns influenced the final race outcome.

3.2 The Dataset

The data was retrieved using the FastF1 Python library [5], extracting timing and telemetry data directly from of-

ficial Formula 1 APIs. To meet the project requirements and maintain a focused analysis, we narrowed the scope to the 2024 Monza GP. The final dataset is structured into four main categories:

- **Driver & Team Info:** Identification data (name, team, color) and final race classification.
- **Performance Metrics:** Quantitative variables including lap times, sector splits (S1, S2, S3), and maximum speeds at the Speed Trap and Finish Line.
- **Lap Context:** Condition-specific attributes such as lap number, track status, and Safety Car deployments.
- **Environmental Factors:** Track and air temperature, and humidity, which are critical for calculating tyre degradation.

The dataset satisfies the required level of complexity, reaching an AS index of approximately 29232. This value is calculated considering about 1,008 rows (derived from roughly 20 drivers over approximately 53 laps) and 29 dimensions. Therefore, the dataset falls well within the recommended complexity range of 10,000–50,000.

3.3 Data Preprocessing

To prepare the data for Visual Analytics and Dimensionality Reduction, several preprocessing steps were implemented:

1. **Feature Aggregation:** Raw lap-by-lap data was aggregated into stints. This allowed us to calculate stint-specific metrics such as the DegradationSlope (the rate of pace loss over time) and average lap times.
2. **Categorical Encoding:** Tyre compounds (SOFT, MEDIUM, HARD) were mapped to numerical values (0, 1, 2) to be processed by the PCA algorithm.
3. **Handling Missing Values:** Missing entries were managed using mean imputation to ensure continuity in the analytical flow.
4. **Standardization:** Features were scaled using a StandardScaler to have a mean of 0 and a variance of 1, preventing variables with larger scales (like top speeds) from dominating the PCA results.

4 Our Solution

To enable technical users to perform an in-depth exploratory analysis, we designed and implemented an interactive dashboard. The system follows the visual analytics mantra: providing an overview of the race while allowing users to drill down into specific stints and technical telemetry through interactive coordination.

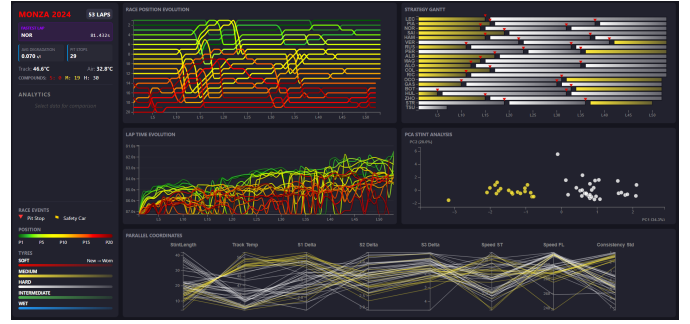


Figure 1: The visual analytics dashboard implemented for race strategy analysis.

4.1 Visual Components

The interface is composed of multiple coordinated views, designed to encode the multidimensional nature of Formula 1 data:

4.1.1 Race Position Evolution

This line chart displays the evolution of the drivers' track positions throughout the race. The x-axis represents the lap number, while the y-axis encodes the race position. Drivers are identified through a continuous color scale ranging from red (representing the driver in last position) to green (representing the race winner). This specific color encoding is consistently applied to the Lap Time Evolution chart, facilitating visual tracking across views.

4.1.2 Lap Time Evolution

This line chart visualizes the pacing of the drivers, plotting the lap number on the x-axis against the registered lap time on the y-axis. A key analytical feature of this view is the dynamic calculation of tyre degradation: when a specific stint is selected by the user, a dashed trend line is overlaid on the chart. This dashed line represents the linear regression of the lap times within that stint, offering an immediate visual quantification of the pace drop-off over time.

4.1.3 Strategy Gantt

This view employs a Gantt chart layout to illustrate the race strategies adopted by the grid. The x-axis maps the laps, while the y-axis lists the drivers, ordered by their final race classification. Each horizontal bar represents a stint (a sequence of consecutive laps driven on the same set of tyres). Pit stops are explicitly encoded using a red reversed triangle marker. The visual variables for the bars strictly follow the standard Formula 1 domain conventions for tyre compounds: Red (Soft), Yellow (Medium), White (Hard), Green (Intermediate), and Blue (Wet). Additionally, the luminance (shadow level) of each bar is used to encode the progressive consumption of the tyre, adding information regarding tyre life.

4.1.4 PCA Stint Analysis

This scatterplot visualizes the results of the Principal Component Analysis (PCA) applied to the stint features. Each point represents a single stint and is positioned according to the first two principal components. Points are color-coded based on the tyre compound used (without the degradation shadow), allowing the user to easily identify clusters of similar driving behaviors or outlier performances.

4.1.5 Parallel Coordinates (PCP)

To inspect the multivariate attributes of the stints, a Parallel Coordinates plot is provided. Each polyline represents a single stint, intersecting vertical axes that encode quantitative metrics: Stint Length, Track Temperature, Sector Deltas (S1, S2, S3), Speed Traps (ST and FL), and Consistency Std (standard deviation of lap times).

4.1.6 Sidebar

The sidebar acts as the control and summary panel, divided into three sections:

- **General Statistics:** provides aggregate context about the entire race, including the fastest lap, average tire degradation, total pit stops, and average environmental temperatures.
- **Analytics (Details-on-Demand):** this section dynamically generates comparative box plots based on user selections. It compares the pace and degradation distribution of the selected stints against the global race baseline.
- **Legend:** clarifies the visual encodings adopted throughout the dashboard (e.g., position colors, tire compounds, and race events).

4.2 User Interaction and Analytics

The system avoids traditional static filters, relying entirely on Brushing and Linking to maintain the analytical flow. Users can brush a cluster of points in the PCA plot or select a stint in the Gantt chart. This action immediately highlights the corresponding data across all other views (e.g., highlighting the specific lap times in the line chart and the technical polylines in the PCP). So, thanks to a seamless brushing and linking implementation, selecting an item in one view immediately highlights the corresponding data across all other visualizations. Crucially, these interactions trigger real-time analytical computations. When a selection is made, the dashboard calculates the linear regression used for the dashed trend line in the Lap Time Evolution chart and computes the statistical distributions for the comparative box plots in the sidebar. This mechanism effectively transforms visual exploration into actionable analytical insights.

5 Insights

This section demonstrates how the dashboard's Visual Analytics capabilities allow for a deep understanding of the 2024 Monza Grand Prix dynamics, moving from macro-strategic narratives to granular technical anomalies.

5.1 Macro-Strategy: The One-Stop Efficiency

The Strategy Gantt chart reveals the core strategic narrative of the race: the winner, Charles Leclerc, adopted a bold one-stop strategy, transitioning from Medium to Hard compounds on lap 15. This approach contrasted sharply with the two-stop strategy of Oscar Piastri, who finished second despite starting from pole position.

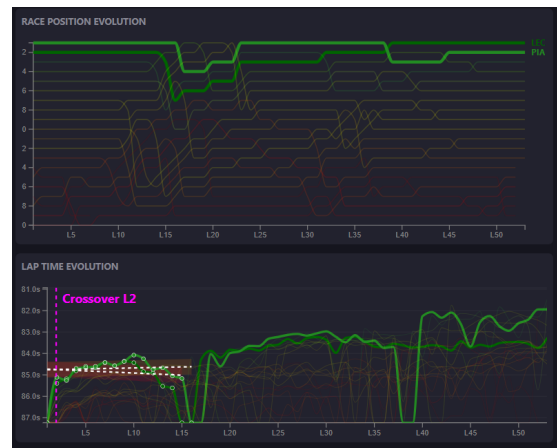


Figure 2: Lap time evolution of Piastri and Leclerc with degradation analysis on laps 1-16

As illustrated in the Line Chart in Figure 2, Piastri was initially faster during the opening stint. Leclerc then executed his pit stop at lap 15, differently from Piastri that stopped to change his tyres at lap 16. Figure 3 shows that during the second stint Piastri maintained a faster pace, but around lap 30 the lap times of the two drivers began to converge. The dashboard's Crossover Point analysis highlights that the performance advantage of Piastri's tyres was no longer sufficient to offset Leclerc's track position advantage. For this reason, Piastri is forced to perform another pit stop at lap 38. By looking at Figure 4, it's possible to observe that while Piastri was pit stopping, Leclerc gained the first position. Then, after the pit stop Piastri's was faster than Leclerc but the race ended before he could overtake him. The box plot represented in Figure 4 shows in fact that Piastri was faster than Leclerc during the race, but the time that Piastri lose during the second pit stop is the main reason of his second position.



Figure 3: Lap time evolution of Piastri and Leclerc for laps 16-38



Figure 4: Lap time evolution of Piastri and Leclerc with degradation analysis on laps 38-53 for Piastri and 16-53 for Leclerc



Figure 5: Box plot comparing Leclerc's lap times in his second stint (laps 16-53) and Piastri's lap times in his final stint (laps 38-53)

5.2 Micro-Technical Insights: PCA Outlier Analysis

The PCA Stint Analysis scatterplot serves as the primary tool for identifying technical anomalies that deviate from standard performance clusters:

1. **Hulkenberg's Initial Stint:** While most Medium-tyre stints are clustered together, Nico Hulkenberg's first stint is a clear outlier. The system identifies this isolation due to a minimum Stint Length (only 5 laps) and significantly inferior performance in sectors S2 and S3, combined with the worst registered Speed FL value.

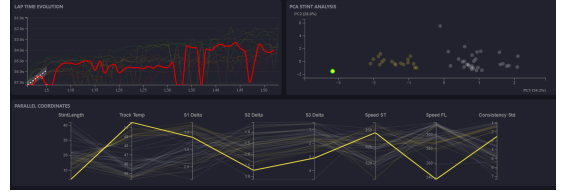


Figure 6: Parallel Coordinates View of Hulkenberg first stint

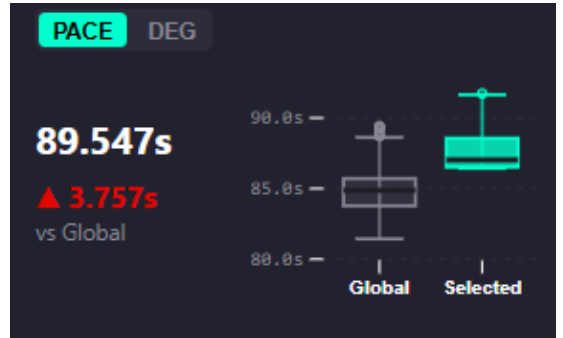


Figure 7: Box plot comparing Hulkneberg lap times over its first stint with the global ones

2. **Tsunoda's Technical Failure:** Another isolated point in the PCA corresponds to Yuki Tsunoda's Hard-tyre stint. The dashboard clarifies the anomaly: it is the only Hard-compound stint lasting only 7 laps. Despite decent Speed ST values, Tsunoda registered the worst sector 2 and 3 performances before his definitive retirement, highlighting a technical failure rather than a strategic choice.

5.3 Human Factor: Driver Ability in Tyre Management

The system demonstrates that final performance depends not only on team strategy but also on the driver's individual ability to manage tyre stress.

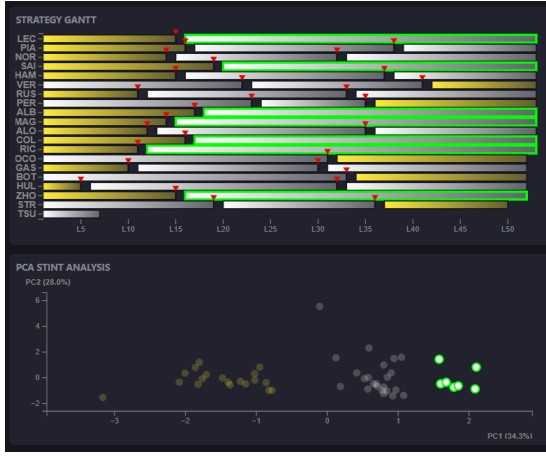


Figure 9: PCA Scatterplot synchronized with Gantt Strategy Chart

- **Cluster Deviations:** By selecting drivers on a one-stop (Medium-Hard) strategy with high lap counts (as done in Figure 9, Ricciardo and Zhou appear isolated from the main leader cluster. This is largely due to both drivers failing to reach the same top speeds (Speed ST) as the leaders within that cluster.
- **Zhou vs. Leclerc Comparison:** Specifically, Zhou Guanyu showed an inability to manage the Hard compound effectively compared to top-tier drivers. He experienced significantly higher degradation, resulting in an Average Lap Time of 86.84s—nearly 2 seconds slower than Leclerc’s average during the same stint. This insight confirms that a driver’s "tyre-whispering" ability is a technical factor as decisive as the strategy itself

6 Conclusion

This project successfully developed a comprehensive Visual Analytics system tailored for the technical analysis of Formula 1 race dynamics. By moving beyond static telemetry and offering an interactive, exploratory environment, the system empowers domain experts to understand the multidimensional factors that determine race outcomes. The successful integration of Principal Component Analysis proved essential for identifying performance clusters and isolating significant outliers, such as the strategic anomalies of Hulkenberg and Tsunoda. Furthermore, the implementation of real-time analytical triggers—including dynamic linear regressions for tyre degradation—ensured that the tool moved beyond simple data visualization into the realm of actionable race intelligence. Future developments could focus on integrating live telemetry streams to support real-time decision-making during a race or extending the system to include fuel-load modeling and meteorological forecasts to further enhance its predictive capabilities.

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